# N83 K899: the next factor-base comparison

*Preparatory source assessment · 26 October 2026*

The current N83 Koblitz prime subgroup has order 8,569,786,107,849,059 (52.93 bits), curve `EC1N83Ce1h35c7ced6b497`. Its current K600 rank-finder producer is run `39527997793`. A nested K899 base was measured only under an older rank-probe producer, run `39306394645`.

The older within-run comparison reports K600/K899 cold 1.1873 [1.1247, 1.2523] and rank-stage 1.5176 [1.4149, 1.6235], over 20 targets and four rank blocks. Construction of either factor base was excluded. K899 had 2.0766 times the non-rank cold work and 2.0713 times peak RSS. The result cannot be stacked arithmetically with later producer gains.

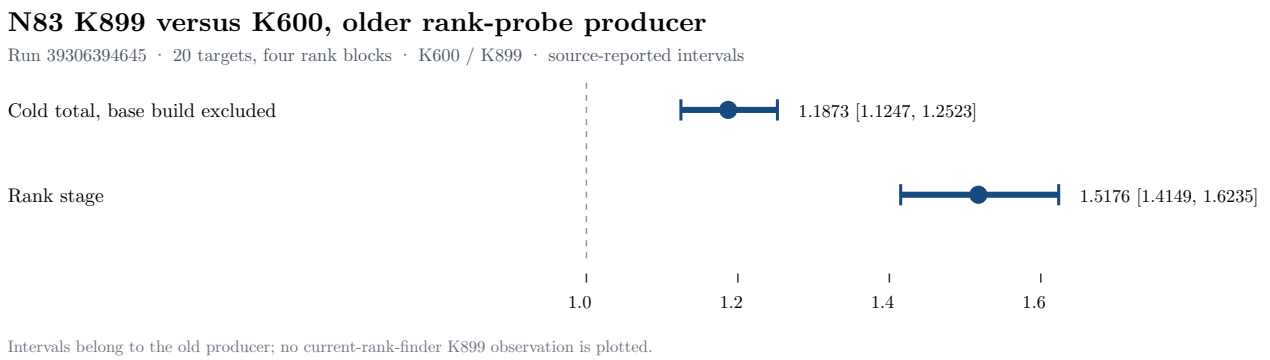


Figure 1: Source-reported paired ratios, run 39306394645. Both ratios exclude factor-base construction. The current rank-finder producer was not an arm of this run.

## Next test boundary

Freeze the current producer and held-out targets. Build both target-blind bases on the same host, charge construction, certification, rank collection, index and linear algebra, online target work and verification; pair rho on the same target and resource envelope. Alternate K600/K899 order and retain A/A, planted, censored, and failed calls. Record useful rank and matrix density as well as wall, CPU and RSS. Separate cold one-target and stated reuse-count totals.

The test needs a frozen `experiment.protocol/v1`, power/stopping and archive plans, preapproval, and Coordinator approval before execution. It answers one Koblitz one-arity question. Prime, ordinary binary and other arities remain open.

## Source and literature

Cryptanalysis `docs/ic/BASELINES.md`, commit `e7cf096b4e2f9020e1326e4ed0b5ac0e96cc99f9`, blob `4a444fac7f5c61992559a57601ee03c48cddab55`. The mechanism was checked against Semaev 2004 (IACR 2004/031), Faugère et al. 2012 (EUROCRYPT), and Huang et al. 2015 (DOI 10.1186/s40736-015-0012-6). Detailed provenance and links are in `README.md`. No new scientific status or quantitative claim is made.